Table 1.1
Comparison of GES- and PC-Derived Causal Edges for the Eclipse Dataset

EDGE	GES	PC	Agreement
CC→ET	✓	✓	Match
OW→ET	✓	✓	Match
OE→ET	✓	✓	Match
RI→ET	✓	-	GES Only
SI→ET	✓	-	GES Only
NI→ET	✓	-	GES Only
SI→CC	✓	✓	Match
RI→CC	✓	✓	Match
FI→CC	✓	✓	Match
NI→CC	✓	✓	Match
Pr→CC	✓	-	GES only
OE→CI	✓	✓	Match
CC→CI	✓	✓	Match
FI→CI	✓	✓	Match
OW→CI	-	✓	PC only

Table 1.2
Comparison of GES- and PC-Derived Causal Edges for the Mozilla Dataset

EDGE	GES	PC	Agreement
CC→ET	✓	✓	Match
OW→ET	✓	✓	Match
OE→ET	✓	✓	Match
RI→ET	✓	-	Match
Pr→ET	-	✓	PC only
RI→CC	✓ (CC→RI)	✓ (RI → CC)	Direction conflict
FI→SI	✓(SI→FI)	✓ (FI→SI)	Direction conflict
CC→SI	✓	✓	Match
Pr→CC	✓(Pr→CC)	(CC→Pr)	Direction conflict
OE→CI	✓	✓	Match
CI→CC	✓	✓	Match
FI→SI	✓(SI→FI)	✓(FI→SI)	Direction conflict

1. DAG using PC algorithm

1.1. Eclipse

1.2. OpenStack

1.3. Mozilla

1.4. Edge Comparison

2. AnchorTabular Consistency Check

An anchor is a minimal set of feature-value conditions under which a trained model's prediction holds with high probability regardless of other feature values. Precision is the probability that the prediction is unchanged under perturbation of non-anchor features, a measure of local prediction stability rather than accuracy against ground truth; coverage is the share of the dataset to which the anchor applies. Anchors are computed only from the fitted predictive model, not from the GES graph or the ATEs, so we use them as a qualitative cross-check on whether the same variables recur, not as a second estimate of causal effect.

We used Alibi's AnchorTabular explainer with a target precision of 0.95 (Table 2.1). Eclipse reaches 0.96 precision at 6% coverage. OpenStack reaches 0.92 at 13% coverage, short of the target. Mozilla required re-discretising into 10-percentile bins to reach 0.96 precision, at which point coverage fell to roughly 1–2 bugs (0.02%). We read the OpenStack anchor as broader but less precise than Eclipse's, and the Mozilla anchor as high-precision but too narrow

Table 2.1
AnchorTabular precision and coverage across ecosystems

Dataset	Anchor Conditions	precision	Coverage	Interpretation
Eclipse	$CC \leq 4.00$; $OE > 108.00$; $RI \leq 6.00$; $955.00 < OW \leq 2882.00$; $SI \leq 3.00$	0.96	0.06	High-confidence rule, applies to 6% of bugs
OpenStack	$CC \leq 3.00$; $OE > 244.00$; $RI \leq 6.00$; $OW > 67.00$; $SI \leq 0.00$; $NI \leq 0.00$	0.92	0.13	Slightly lower confidence, broader applicability than Eclipse or Mozilla
Mozilla	$CC > 4.00$; $8.00 < RI \leq 11.00$; $6.00 < SI \leq 10.00$; $OE \leq 993.82$; $OW = 0.00$; $3.00 < CI \leq 6.00$; $3.00 < Pr \leq 4.00$	0.87 → 0.96	0.0071 → 0.0002*	*Initial 4-bin discretisation gave precision 0.87; refining to 10-percentile bins raised precision to 0.96 but reduced coverage to 0.02% (1–2 bugs; see caveat below.

to generalise beyond the handful of bugs it covers; we treat it as weak corroborating evidence rather than a consistency check on the same footing as the other two.

Contributors (CC), OwnerWorkload (OW), RepeatInteraction (RI) and SeasonedInteraction (SI) appear as anchor conditions in all three ecosystems (Table 2.1); many of them have their role as GES parents of ET. Anchor membership indicates only that the model's prediction is sensitive to these variables in this region of the feature space, not the direction or size of their effect, and it describes the same predictive model that Causal SHAP weights rather than an independent source of evidence. We treat the recurrence of CC, OW, RI and SI across GES, Causal SHAP and AnchorTabular as one consistency check with three components, not as three separate confirmations.

2.0.1. Per-ecosystem anchor rules

As model complexity increases, it becomes difficult to describe the model's decisions. Anchors reduce the complexity of the explanation by distributing accurate, rule-based explanations that are intended for certain chosen predictions These explanations help stakeholders and users understand the reasons behind the model's decision. Anchors represent concepts used to search for base guidelines that are sufficiently informative to 'anchor' a model's prediction to specific features of the data.

Table 2.1 reports the anchor rules found for each ecosystem using Alibi's AnchorTabular explainer, with a target precision threshold of 0.95. An anchor is a minimal set of feature-value conditions sufficient to keep the model's prediction unchanged with high probability, regardless of the values of other features; precision is the empirical probability that the prediction holds under perturbation of non-anchor features, and coverage is the proportion of the dataset to which the anchor conditions apply.

2.0.2. Cross-ecosystem precision

*For Mozilla, quantile-based discretisation into four bins was initially too coarse to reach the 0.95 precision target (0.87 achieved). Re-discretising at 10-percentile bins raised precision to 0.96, but reduced coverage to 0.02% (roughly 1–2 bugs). We treat this narrow, high-precision Mozilla anchor as weak corroborating evidence — it risks describing an idiosyncratic subpopulation rather than a generalisable pattern — rather than as a consistency check on the same footing as the Eclipse and OpenStack anchors.

3. Edge stability under resampling

3.1. Procedure.

To assess how sensitive the learned structures are to sampling variation, we drew 200 bootstrap resamples (rows sampled with replacement, same size as the original sample, seeds 0–199) from each dataset, standardised each

Table 3.1
Edge stability for the parents of ET: percentage of 200 bootstrap resamples (GES, BIC, $\lambda = 0.5$, standardised data) in which the pair is adjacent, and in which the edge is oriented $X \to \mathrm{ET}$ or $\mathrm{ET} \to X$.

Dataset	Edge	Adjacent	$X \to \mathrm{ET}$	$\mathrm{ET} \to X$
Eclipse	OW → ET	100	100	0
	OE → ET	100	100	0
	CC → ET	100	87	13
	RI → ET	91	82	10
	SI → ET	89	67	22
	NI → ET	69	60	9
OpenStack	CC → ET	100	100	0
	OW → ET	100	100	0
	RI → ET	100	100	0
	Pr → ET	100	100	0
	OE – ET	99	44	52
Mozilla	CC → ET	100	42	50
	OW → ET	48	4	44
	OE → ET	100	17	42
	RI → ET	85	23	59

resample, and re-ran GES with the settings of (BIC score, $\lambda = 0.5$). For each pair of variables we recorded how often the pair was adjacent in the learned graph and, if adjacent, how often the edge was oriented each way or left undirected. The bootstrap measures stability under resampling; it is not a significance test, and edge frequencies are not probabilities that an edge is causal.

3.2. Result.

Table 3.1 gives the frequencies for the parents of ET. In OpenStack, CC, OW, RI and Pr were adjacent to ET and oriented into it in all 200 resamples. The OE–ET edge was present in 99% of resamples but its direction was not stable (ET→OE 52%, OE→ET 44%), which agrees with the direction conflict with PC. In Eclipse, OW→ET and OE→ET were returned in every resample and CC→ET in 87%; RI→ET, SI→ET and NI→ET were less stable (82%, 67% and 60%). In Mozilla, CC and OE were adjacent to ET in every resample, but the orientation varied, and OW and RI were adjacent in fewer resamples. Repeating the bootstrap at $\lambda = 1$ and $\lambda = 2$ gave sparser graphs. The edges to ET that were stable at $\lambda = 0.5$ remained so, whereas the Eclipse SI and NI edges and the Mozilla OW and RI edges lost adjacency as λ increased. A larger penalty therefore does not stabilise these edges; it removes them.

3.3. Interpretation.

The results support using the OpenStack structure and the Eclipse OW, OE and CC edges as the more reliable ones, and they argue for caution with the interaction-count edges in Eclipse and with all Mozilla edges. They also bear on the cross-ecosystem comparison: where an edge appears in one ecosystem but not another (for example SI→ET), the difference is within the range of sampling variation in the weaker graph. Because the resamples come from the same observations, this check does not address model misspecification, omitted confounders, or the heavy-tailed variables.

Table 3.2
Edge stability of the edges into ET at three penalty weights λ. Each cell gives the percentage of 200 bootstrap resamples in which the pair is adjacent, the edge is oriented $X \to \mathrm{ET}$, and the edge is oriented $\mathrm{ET} \to X$ (GES, BIC, standardised data). The remaining resamples left the edge undirected.

Dataset	Edge	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$
		adj / $X \to$ET / ET$\to X$	adj / $X \to$ET / ET$\to X$	adj / $X \to$ET / ET$\to X$
Eclipse	CC → ET	100 / 87 / 13	100 / 87 / 13	100 / 87 / 13
	OW → ET	100 / 100 / 0	100 / 100 / 0	100 / 100 / 0
	OE → ET	100 / 100 / 0	100 / 100 / 0	100 / 100 / 0
	RI → ET	91 / 82 / 10	86 / 78 / 8	62 / 58 / 4
	SI → ET	89 / 67 / 22	68 / 50 / 18	40 / 29 / 10
	NI → ET	69 / 60 / 8	49 / 42 / 6	34 / 30 / 4
OpenStack	CC → ET	100 / 100 / 0	100 / 100 / 0	100 / 100 / 0
	OW → ET	100 / 100 / 0	100 / 100 / 0	100 / 100 / 0
	RI → ET	100 / 100 / 0	100 / 100 / 0	99 / 99 / 0
	Pr → ET	100 / 100 / 0	100 / 100 / 0	86 / 86 / 0
	OE – ET	99 / 44 / 52	74 / 40 / 34	2 / 1 / 2
Mozilla	CC → ET	100 / 42 / 50	100 / 44 / 52	98 / 38 / 52
	OW → ET	48 / 4 / 44	24 / 3 / 22	11 / 2 / 10
	OE → ET	100 / 16 / 42	100 / 14 / 34	100 / 8 / 30
	RI → ET	84 / 23 / 59	72 / 20 / 50	17 / 2 / 15

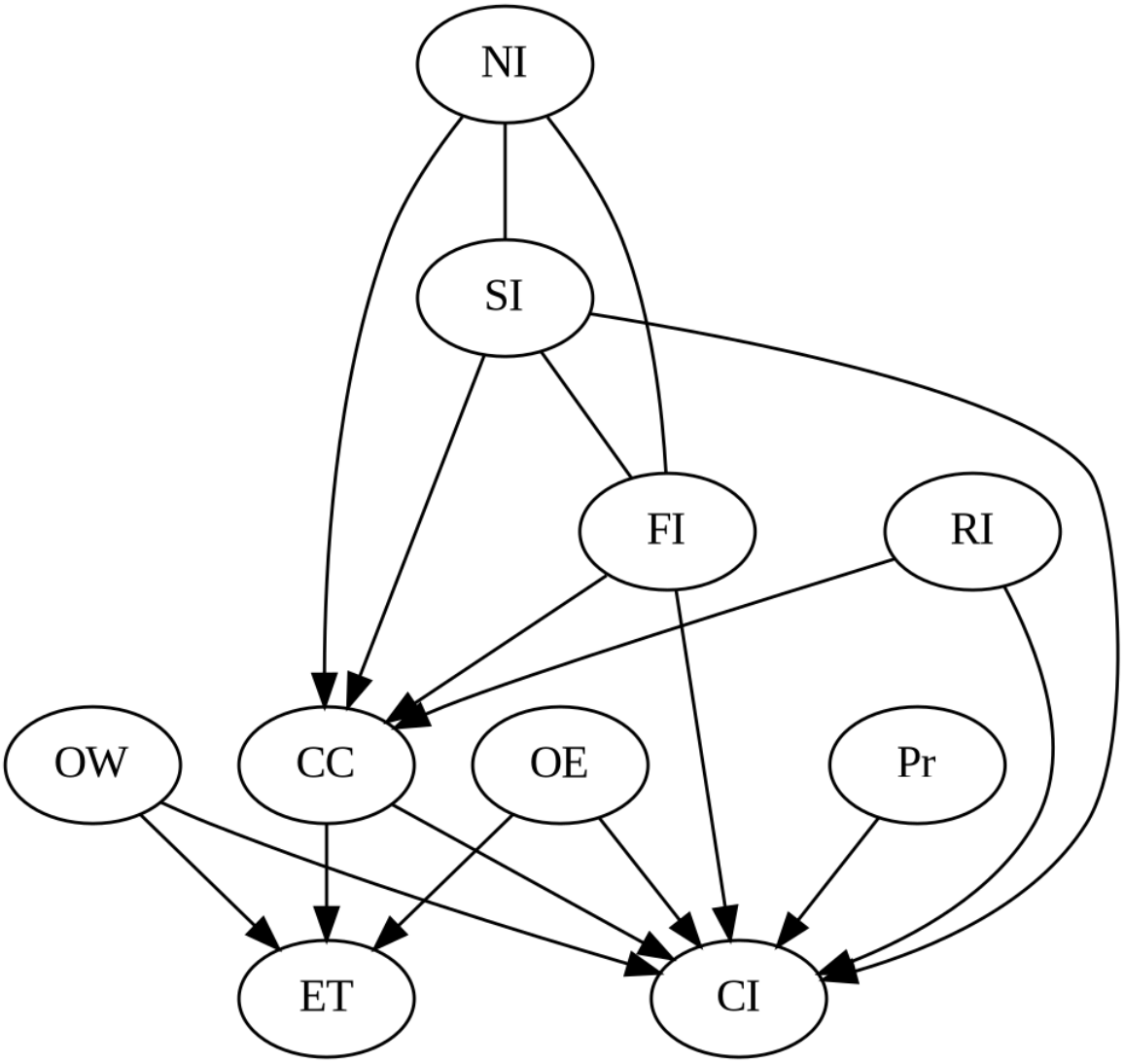


Figure 1.1: DAG for Eclipse

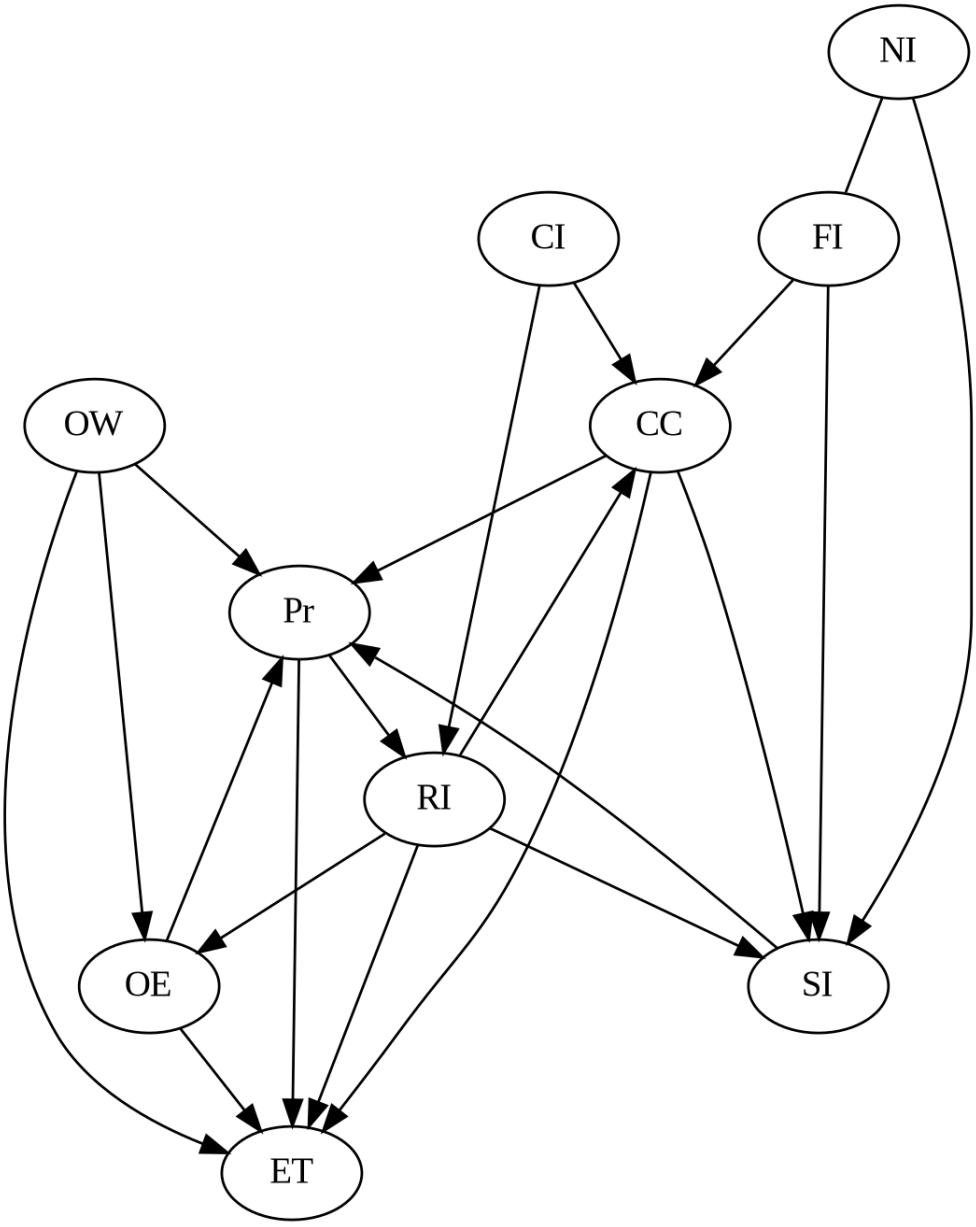


Figure 1.2: DAG for OpenStack

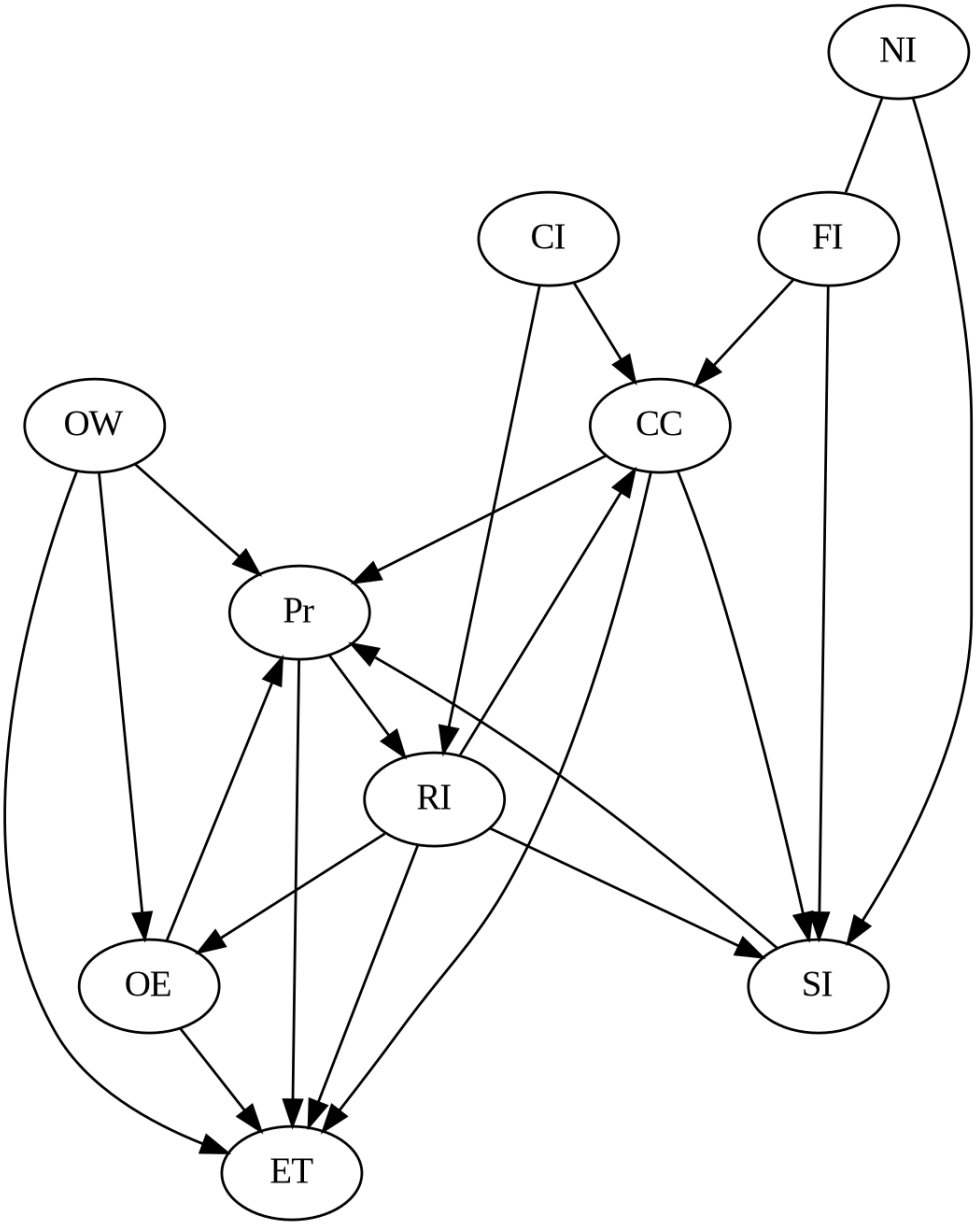


Figure 1.3: DAG for Mozilla